Miracle Agbebaku

AI and agent systems engineer

miracleagbebaku122@gmail.com · WhatsApp +234 902 163 7905 · miracle-agbebaku-portfolio.vercel.app

SUMMARY

Builds AI-agent systems where the model proposes and deterministic code decides: structured plans, fail-closed policy checks, risk-tiered permissions, audit trails and tests. Three systems so far: an agent governance pipeline, a WhatsApp assistant with provider fallback and an in-app assistant with sandboxed code execution.

PROJECTS

NEXUS CORE

Working prototype

Next.js, React, TypeScript, FastAPI, Pydantic, SQLite

Governance-first agent orchestration: the model plans, a deterministic fail-closed policy engine decides.

- Built a planner and a correction agent whose structured plans are validated with Pydantic and checked by a deterministic, fail-closed policy engine before anything runs. Blocked plans get at most two bounded corrections, each checked again.
- Streams each pipeline stage over Server-Sent Events and records SHA-256 hash-linked audit events, so every decision can be inspected afterwards.
- Kept execution honest: mostly simulated, with read-only web search as the one real tool, and the limits written down.

ELLA

Active development

TypeScript, Node.js, Express, Supabase, React, Vite

WhatsApp assistant built around NEXUS CORE, running many linked accounts in one process.

- Added a risk-tiered permission engine (five roles, four risk levels) in which critical tools need approval, and a response guard that strips internal data before every outgoing message.
- Built a provider gateway with ordered fallback and health tracking, plus rules that treat links and fetched pages as hostile input in the media and research tools.
- Handles Nigerian English and Pidgin, and is covered by about 50 test scripts.

Lagoon

In development

JavaScript, Cloudflare Workers, Durable Objects, WebSockets, Playwright

Social platform with an in-app assistant and two code-execution sandboxes.

- Built the assistant in layers: local reasoning first, server AI second, optional user-configured providers third, with a fixed action allowlist and model output rendered as data, not markup.
- Runs code in two separate sandboxes: an emulated Linux environment inside the browser and a remote runner that accepts only signed requests, with limits on code size, output and execution time.
- Used fixed-host data adapters so a user-supplied address cannot become an unrestricted server fetch.

SKILLS

Languages	Python, TypeScript, JavaScript
Backend and APIs	FastAPI, Pydantic, Node.js, Express, Zod
Realtime and edge	Cloudflare Workers, Durable Objects, WebSockets, Server-sent events
Data	SQLite, Supabase, PostgreSQL
Testing and delivery	Playwright, Vitest, GitHub Actions, Git, Render, Vercel
Integrations	WhatsApp, FFmpeg
Practices	policy gates, audit logging, input validation, sandboxing, automated testing

WRITING

20 engineering notes on trust boundaries, AI agent pipelines, messaging systems and how the portfolio is built, at miracle-agbebaku-portfolio.vercel.app/engineering.